

Security Metrics & Data Analysis Report

1. Executive Summary

This report examines global cybersecurity incidents from 2015 to 2024 to understand attack trends, the industries most at risk, financial losses, and how organizations respond to security incidents. The goal is to provide useful insights that can help organizations strengthen their cybersecurity and make better decisions when planning and investing in security measures.

2. Business Problem Overview

Cyberattacks are becoming more frequent and more advanced, creating serious challenges for organizations around the world. This report uses data from global cybersecurity incidents between 2015 and 2024 to understand attack patterns, financial impacts, vulnerabilities, and the effectiveness of security measures. The findings help organizations identify risks and make better decisions to improve cybersecurity.

▪ Most Common Types of Attacks

Phishing, Distributed Denial-of-Service (DDoS), and ransomware are the most common cyberattacks across different countries and industries. For example, in 2019, a phishing attack in China's education sector affected 773,169 users. This shows that these attacks continue to be major threats to organizations and individuals.

▪ High-Risk Industries

The healthcare and finance sectors experience the highest number of affected users and financial losses. For example, a DDoS attack on France's healthcare sector in 2016 affected 996,004 users. Organizations that store sensitive personal and financial information are attractive targets for cybercriminals and need stronger security measures.

▪ High Financial Risk

Cyberattacks can cause very large financial losses. Ransomware and DDoS attacks sometimes result in losses of tens of millions of dollars in a single incident. For example, Germany's banking sector lost \$87.56 million from a ransomware attack in 2018. These incidents show that cybersecurity is not only a technical issue but also a major business and financial risk.

▪ Critical Vulnerabilities

Weak passwords and unpatched software are the most common weaknesses used by attackers. Although these are basic security problems, many organizations still fail to fix them. This suggests that companies need stronger cybersecurity practices, regular software updates, and better security awareness.

- **Nation-State and Insider Threats**

Attacks carried out by nation-state groups caused some of the highest financial losses and often used advanced methods such as zero-day exploits. Insider threats also make attacks more difficult to detect and respond to. These threats increase the complexity of cybersecurity incidents and create risks that go beyond normal business operations.

- **Effectiveness of Current Defenses**

The data shows that AI-based detection systems can significantly improve incident response times. Organizations using these systems often resolved incidents in less than 20 hours. In contrast, traditional security tools such as firewalls and antivirus software often required 50 to 70 hours to resolve incidents. This suggests that investing in advanced security technologies can improve cyber defense and reduce business impact.

Business Problem Statement

Organizations face increasing cybersecurity risks that threaten financial stability, customer trust, and business operations. Cyberattacks continue to exploit basic security weaknesses, target critical industries, and overcome traditional security measures. Therefore, businesses need data-driven strategies to identify attack patterns, strengthen weak areas, improve response times, and invest in smarter security technologies to reduce risks and maintain operational continuity.

3. Project Objectives

The main goal of this project is to analyze global cybersecurity incidents from 2015 to 2024 and provide useful insights that can help organizations improve their cybersecurity and reduce risks.

1. Attack Landscape Analysis

Objective: Identify the most and least common cyberattacks and examine how attacks vary by year and country.

Business Value: Helps organizations understand common threats and focus security resources on high-risk regions.

2. Industry Risk Analysis

Objective: Identify which industries are targeted the most and which suffer the highest financial losses and user impact.

Business Value: Helps industries such as healthcare and finance develop stronger security strategies.

3. Financial Impact Analysis

Objective: Measure the total and average financial losses caused by different types of attack, countries, and industries.

Business Value: Supports better risk management and helps organizations decide where to invest in cybersecurity.

4. User Impact Analysis

Objective: Analyze the number of users affected by cyberattacks and identify the most damaging attacks.

Business Value: Helps organizations understand the impact on customers and improve communication and incident response plans.

5. Vulnerability Analysis

Objective: Identify the most common vulnerabilities used by attackers.

Business Value: Encourages better patch management and helps organizations fix weak areas in their systems.

6. Attack Source Analysis

Objective: Examine different attack sources, such as hacker groups, nation-state actors, and insiders, and compare their financial impact and resolution times.

Business Value: Helps organizations develop security measures against different types of attackers.

7. Defense Mechanism Analysis

Objective: Evaluate the effectiveness of different security controls and compare their resolution times and financial losses.

Business Value: Helps organizations invest in the most effective cybersecurity technologies.

8. Incident Response Analysis

Objective: Analyze how long organizations take to resolve incidents based on attack type, industry, and country.

Business Value: Identifies weaknesses in incident response processes and improves security operations.

9. Trend Analysis (2015–2024)

Objective: Study long-term trends in cyberattacks, financial losses, vulnerabilities, and response times.

Business Value: Helps organizations predict future risks and plan long-term cybersecurity strategies.

10. Correlation Analysis

Objective: Explore the relationships between financial losses, affected users, and resolution times.

Business Value: Reveals important patterns that can improve overall risk management and decision-making.

Overall Project Objective

The overall objective of this project is to use historical cybersecurity data to understand attack trends, industry risks, financial and user impacts, and the effectiveness of security measures. The insights from this analysis can help organizations strengthen their cybersecurity, improve incident response, and reduce future risks.

4. Dataset Description

This project uses the **Global Cybersecurity Threats (2015–2024)** dataset. The dataset contains information about cybersecurity incidents that happened in different countries and industries between 2015 and 2024.

It includes details about attack types, financial losses, affected users, security weaknesses, defense methods, and incident response times. The dataset helps us understand global cybersecurity trends and the impact of cyberattacks on organizations.

The dataset contains **3,000 records** and **10 variables**. Each record represents one cybersecurity incident.

Dataset Source

- **Source:** Kaggle
- **Dataset Name:** Global Cybersecurity Threats (2015–2024)
- **File Format:** CSV (Comma-Separated Values)

Dataset Size

- **Number of Rows:** 3,000
- **Number of Columns:** 10
- **Time Period:** 2015–2024

Variable Description

Variable	Data Type	Description
Country	Categorical	The country where the cybersecurity incident happened.
Year	Numerical	The year when the incident occurred.
Attack Type	Categorical	The type of cyberattack, such as phishing, malware, ransomware, or DDoS.
Target Industry	Categorical	The industry was affected by attacks such as Healthcare, Finance, or Government.
Financial Loss (in Million \$)	Numerical	The estimated financial loss caused by the incident, measured in millions of US dollars.
Number of Affected Users	Numerical	The number of users affected by the cyberattack.
Attack Source	Categorical	The source of the attack, such as hacker groups, insiders, or nation-state actors.
Security Vulnerability Type	Categorical	The security weakness that was used by the attackers.
Defense Mechanism Used	Categorical	The security method used to detect or stop the attack.
Incident Resolution Time (in Hours)	Numerical	The number of hours needed to contain and resolve the incident.

Purpose of the Dataset

This dataset can be used to study global cybersecurity trends and create security reports and dashboards. It helps answer important business questions, such as:

- Which cyberattacks are the most common?
- Which countries and industries are at the highest risk?
- How much financial damage does cyberattack cause?
- How many users are affected by different attacks?
- Which security measures work best?
- How quickly do organizations respond to cyber incidents?

The insights from this dataset can help organizations improve their cybersecurity strategies, make better decisions, and reduce future risks.

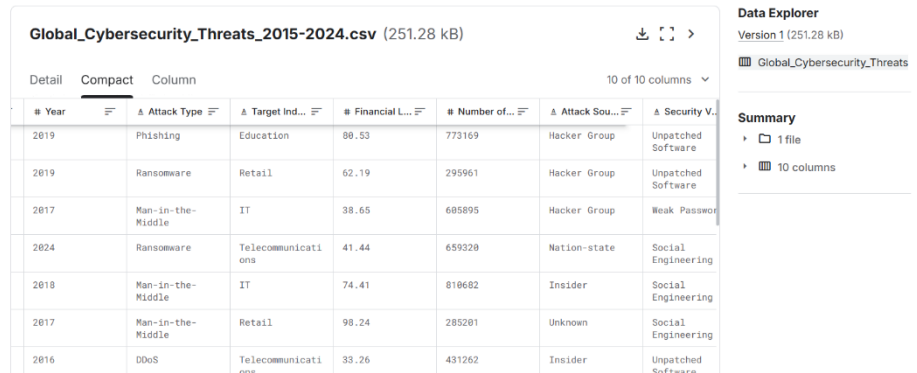
5) Methodology

5.1 Data Collection

The data used in this study comes from the Global Cybersecurity Threats (2015–2024) dataset, which was created and shared publicly for research purposes. This dataset brings together information about real cybersecurity incidents reported across different countries and industries.

It includes details such as:

- The year and type of attack (e.g., phishing, ransomware, DDoS).
- The industry targeted (such as education, retail, IT, telecommunications).
- The financial losses and number of incidents or affected users.
- The attack source (hacker groups, insiders, nation-states, etc.).
- The security vulnerabilities exploited and the defense mechanisms used.



# Year	# Attack Type	# Target Ind...	# Financial L...	# Number of...	# Attack Sou...	# Security V...
2019	Phishing	Education	88.53	773169	Hacker Group	Unpatched Software
2019	Ransomware	Retail	62.19	295961	Hacker Group	Unpatched Software
2017	Man-in-the-Middle	IT	38.65	685895	Hacker Group	Weak Passwords
2024	Ransomware	Telecommunications	41.44	659328	Nation-state	Social Engineering
2018	Man-in-the-Middle	IT	74.41	818682	Insider	Social Engineering
2017	Man-in-the-Middle	Retail	98.24	285281	Unknown	Social Engineering
2016	DDoS	Telecommunications	33.26	431262	Insider	Unpatched Software

Because the dataset is licensed under CC0 (Public Domain), it is freely available for analysis and can be used to study global cybersecurity patterns, forecast future threats, and support machine learning models for digital security.

5.2 Data Cleaning

The dataset was obtained in a pre-cleaned format from the original source, meaning that common issues such as missing values, duplicates, and inconsistent data types had already been addressed. To ensure reliability, several validation steps were carried out:

```
print(df.isnull().sum()) #check for missing value
```

```
Country      0
Year         0
Attack Type  0
Target Industry  0
Financial Loss (in Million $)  0
Number of Affected Users  0
Attack Source  0
Security Vulnerability Type  0
Defense Mechanism Used  0
Incident Resolution Time (in Hours)  0
dtype: int64
```

- Missing values check: All columns contained complete data, with zero null entries.
- Data type verification: Each column was stored in the correct format (e.g., integers for year and affected users, floats for financial loss, categorical values for attack type and country).
- Duplicate records: No duplicate rows were found in the dataset.
- Summary statistics: Descriptive statistics confirmed that values were within expected ranges, with financial loss, affected users, and resolution times showing realistic distributions.

Since the dataset was already cleaned and validated, no additional preprocessing was required, allowing the analysis to proceed directly.

```
[13]: print(df.dtypes) #check data types
```

Country	object
Year	int64
Attack Type	object
Target Industry	object
Financial Loss (in Million \$)	float64
Number of Affected Users	int64
Attack Source	object
Security Vulnerability Type	object
Defense Mechanism Used	object
Incident Resolution Time (in Hours)	int64
dtype:	object

```
print(df.duplicated().sum()) #check for duplicates
```

0

```
print(df.describe())
```

	Year	Financial Loss (in Million \$)	Number of Affected Users	\
count	3000.000000	3000.000000	3000.000000	
mean	2019.570333	50.492970	504684.136333	
std	2.857932	28.791415	289944.084972	
min	2015.000000	0.500000	424.000000	
25%	2017.000000	25.757500	255805.250000	
50%	2020.000000	50.795000	504513.000000	
75%	2022.000000	75.630000	758088.500000	
max	2024.000000	99.990000	999635.000000	

	Incident Resolution Time (in Hours)
count	3000.000000
mean	36.476000
std	20.570768
min	1.000000
25%	19.000000
50%	37.000000
75%	55.000000
max	72.000000

```
df.head() #inspect first few rows
```

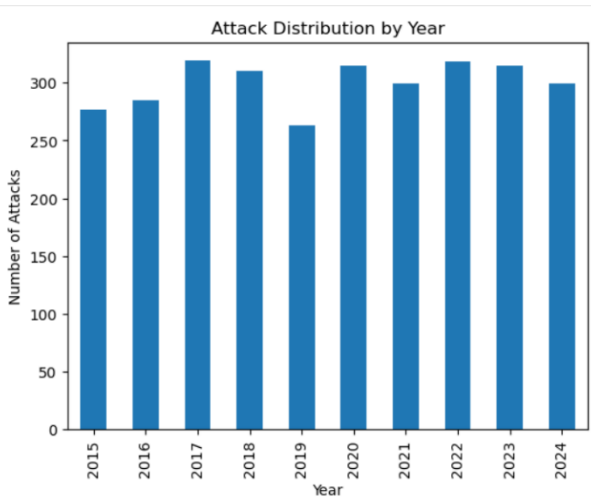
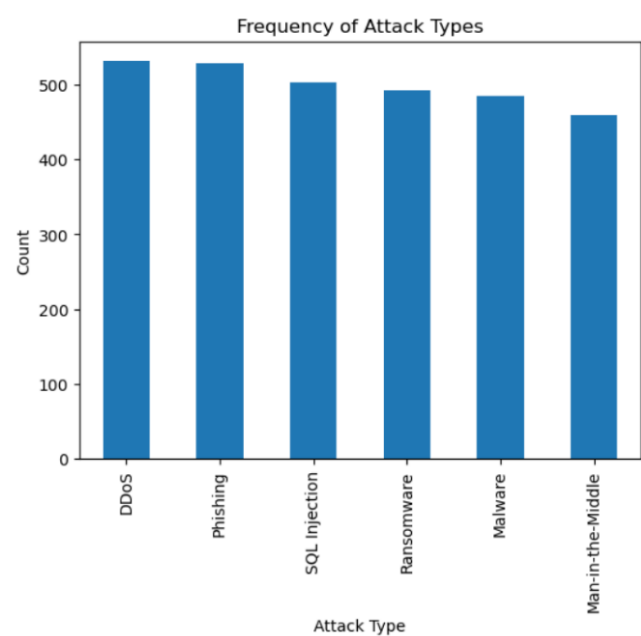
	Country	Year	Attack Type	Target Industry	Financial Loss (in Million \$)	Number of Affected Users	Attack Source	Security Vulnerability Type	Defense Mechanism Used	Incident Resolution Time (in Hours)
0	China	2019	Phishing	Education	80.53	773169	Hacker Group	Unpatched Software	VPN	63
1	China	2019	Ransomware	Retail	62.19	295961	Hacker Group	Unpatched Software	Firewall	71
2	India	2017	Man-in-the-Middle	IT	38.65	605895	Hacker Group	Weak Passwords	VPN	20
3	UK	2024	Ransomware	Telecommunications	41.44	659320	Nation-state	Social Engineering	AI-based Detection	7
4	Germany	2018	Man-in-the-Middle	IT	74.41	810682	Insider	Social Engineering	VPN	68

5.3 Exploratory Data Analysis (EDA)

5.3.1 Attack Landscape Analysis

Analysis Performed

This analysis examined the overall attack landscape by identifying the most and least common cyberattack types. It also analyzed how attacks were distributed across different years (2015–2024) and countries to understand attack patterns and trends.



Visualizations

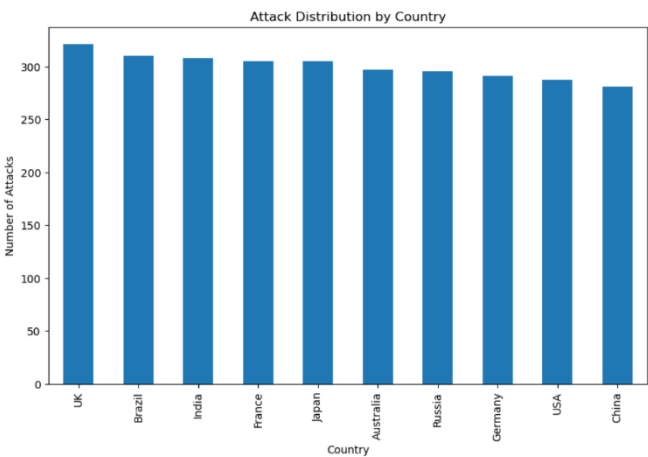
The following visualizations were created to support the analysis:

Frequency of Attack Types

A bar chart showing the number of incidents for each attack type, including DDoS, Phishing, SQL Injection, Ransomware, Malware, and Man-in-the-Middle attacks.

Attack Distribution by Year

A bar chart showing the number of cybersecurity incidents reported each year from 2015 to 2024.



Attack Distribution by Country

A bar chart comparing the number of incidents across countries, including the UK, Brazil, India, France, Japan, Australia, Russia, Germany, the USA, and China.

Results

The analysis shows that **DDoS and Phishing** are the most common cyberattack types in the dataset. These attacks account for a large percentage of all reported incidents, indicating that they remain major threats to organizations worldwide.

In contrast, **Man-in-the-Middle attacks** are the least common attack type. They occur much less frequently than other types of cyberattacks.

The yearly analysis shows that the number of attacks increased during **2017–2018** and again during **2022–2023**. However, there was a noticeable decrease in the number of incidents reported in **2019**.

The country-level analysis reveals that the **United Kingdom (UK)** experienced the highest number of reported cyberattacks in the dataset. On the other hand, **China** reported fewer incidents compared to the other countries included in the analysis.

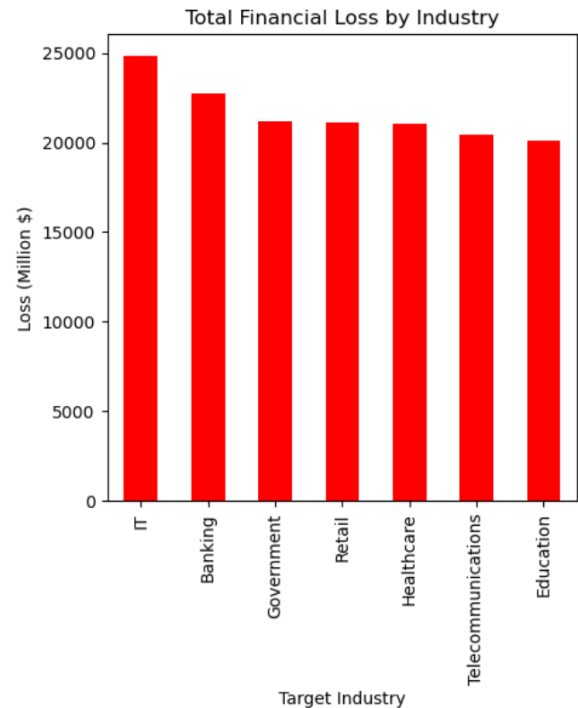
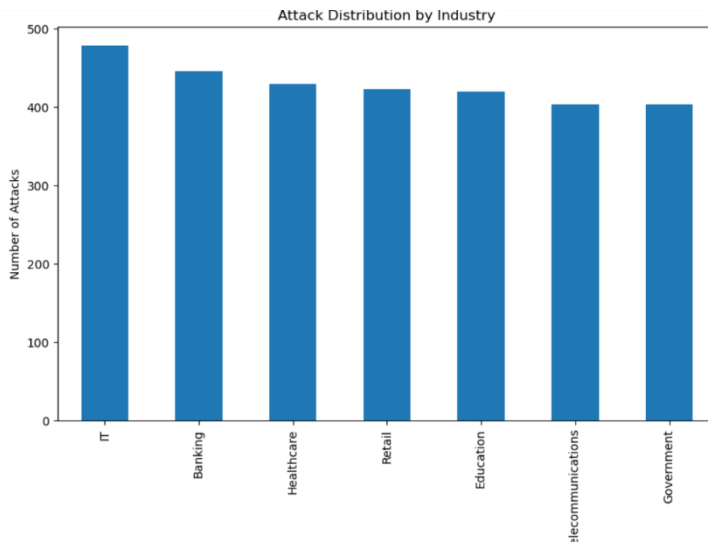
Insight

The results suggest that organizations should pay particular attention to protecting against DDoS and phishing attacks, as these are the most common threats. The increase in attacks during certain years also highlights the need for continuous monitoring and improved cybersecurity measures. Countries and organizations with high numbers of incidents may require additional investment in cybersecurity awareness, detection, and response capabilities.

5.3.2 Industry Risk Analysis

Analysis Performed

This analysis examined the level of cybersecurity risk across different industries. The analysis included counting the number of attacks in each industry, calculating the total financial losses, and measuring the total number of affected users.



Visualizations

The following visualizations were created to support the analysis:

Attack Distribution by Industry

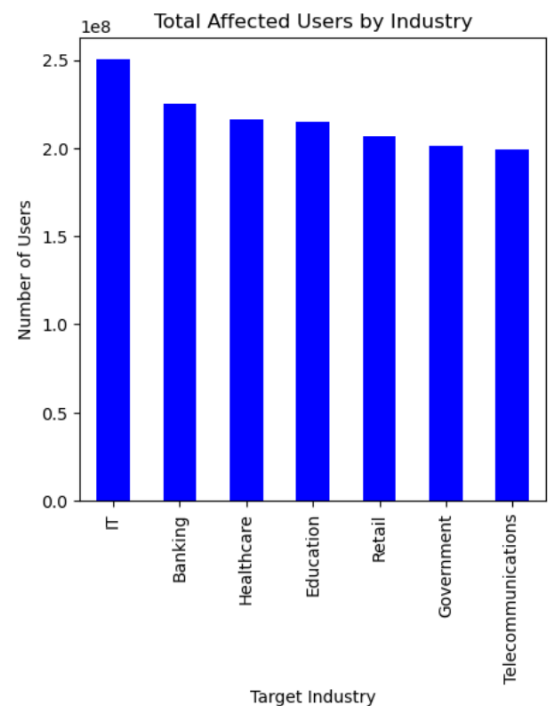
A bar chart showing the number of cyberattacks across industries such as IT, Banking, Healthcare, Retail, Education, Telecommunications, and Government.

Total Financial Loss by Industry

A bar chart comparing the total financial losses (in millions of dollars) experienced by each industry.

Total Affected Users by Industry

A bar chart shows the total number of users affected by cyberattacks in each industry.



Results

The analysis shows that the **IT industry** is the most targeted sector, with nearly **490 reported incidents**. This makes it the industry most frequently attacked by cybercriminals.

The IT sector also experienced the **highest financial losses**, with total losses of approximately **\$25,000 million**. The **Banking** and **Government** sectors followed closely, indicating that organizations holding valuable financial and administrative information face significant cyber risks.

In terms of user impact, the **IT industry** again recorded the highest numbers, with around **250 million affected users**. The **Banking** and **Healthcare** sectors also reported large numbers of affected users, showing that cyberattacks can impact millions of individuals.

Although **Education** and **Telecommunications** experienced fewer attacks and lower financial losses, they still reported a considerable number of affected users. This suggests that even industries with fewer incidents can face significant consequences when attacks occur.

Insight

The industry risk analysis highlights that the **IT sector is the most vulnerable industry**, experiencing the highest number of attacks, the greatest financial losses, and the largest impact on users. This is likely because the IT industry plays a critical role in supporting digital systems and services.

The **Banking** and **Government** sectors also face high levels of risk due to the sensitive financial and administrative data they manage. Cyberattacks against these industries can result in substantial financial damage and operational disruption.

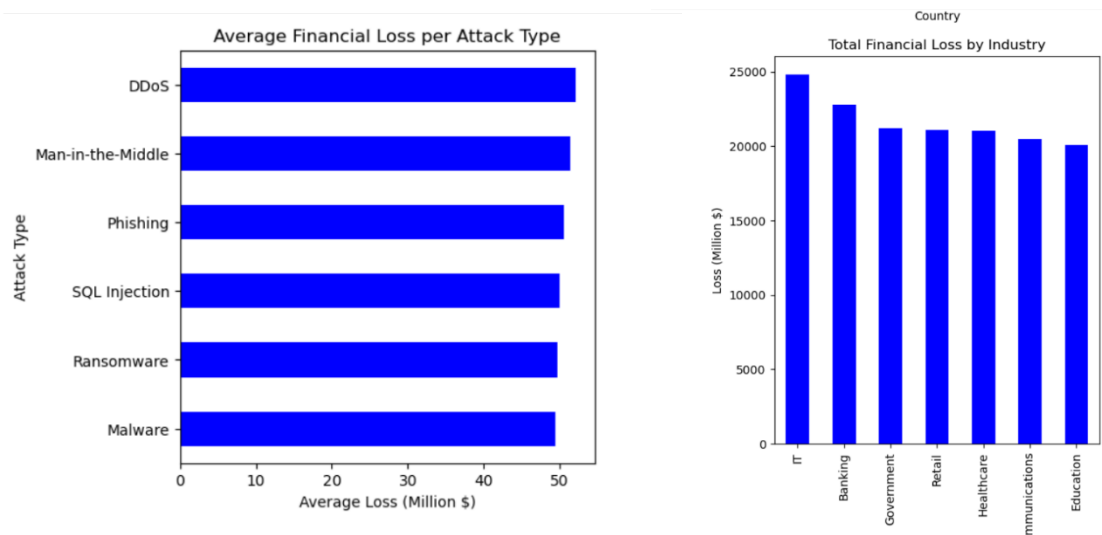
While the **Healthcare** sector experiences fewer attacks than the IT industry, it still shows a high level of user impact. This emphasizes the importance of protecting patient information and strengthening cybersecurity measures within healthcare organizations.

Overall, the findings suggest that industries handling large amounts of sensitive data, such as **IT, Banking, Healthcare, and Government**, face the greatest cybersecurity risks and should be a priority for security investments and risk management efforts.

5.3.3 Financial Impact Analysis

Analysis Performed

This analysis examined the financial impact of cybersecurity incidents between 2015 and 2024. The analysis included calculating the total financial loss across all incidents, measuring the average loss for each attack type, and comparing financial losses across industries and countries.



Visualizations

The following visualizations were created to support the analysis:

Average Financial Loss per Attack Type

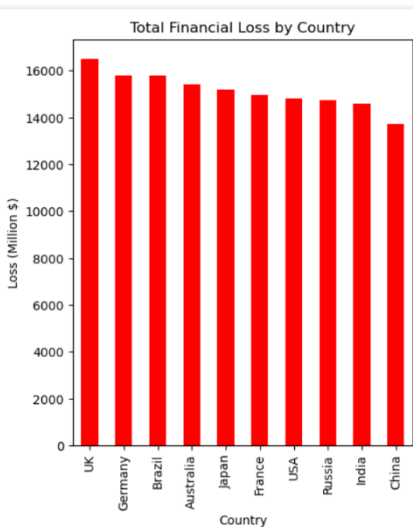
A bar chart showing the average financial loss caused by different attack types, including DDoS, Phishing, Ransomware, Malware, SQL Injection, and Man-in-the-Middle attacks.

Total Financial Loss by Industry

A bar chart comparing the total financial losses across industries such as IT, Banking, Government, Retail, Healthcare, Telecommunications, and Education.

Total Financial Loss by Country

A bar chart showing the total financial losses experienced by different countries.



Results

The analysis shows that Ransomware causes the highest average financial loss per incident. Although phishing attacks occur more frequently, their average financial impact is lower. This indicates that ransomware attacks are more costly and disruptive for organizations.

At the industry level, the IT sector experienced the highest total financial loss, with losses of approximately \$25,000 million. The Banking and Government sectors also recorded significant losses, while the Education sector reported the lowest financial impact.

When comparing countries, the United Kingdom (UK) reported the highest total financial losses, exceeding \$16,000 million. In contrast, China recorded the lowest financial losses among the major countries included in the analysis.

The results show that the largest financial losses are concentrated in industries and countries that depend heavily on digital systems and manage valuable or sensitive information.

Insight

The financial impact analysis highlights the serious economic consequences of cyberattacks. Ransomware is the most expensive attack type because it can disrupt operations, require costly recovery efforts, and lead to significant business losses.

The IT industry faces the greatest financial risk because it supports critical digital infrastructure and services. Similarly, the Banking and Government sectors experience high losses due to the sensitive information they manage and their importance to daily operations.

At the country level, the high losses reported in the UK suggest that countries with advanced digital infrastructure may face greater financial exposure to cyber threats.

Overall, the findings show that cyberattacks can create significant financial damage, especially in industries and countries that rely heavily on technology. Organizations should invest in strong cybersecurity measures and proactive risk management to reduce potential losses.

Financial Loss Summary

Category	Highest Loss Area	Total Loss (Million \$)	Avg. Loss per Incident (Million \$)	Number of Incidents
Overall	All Cyber Attacks	151,478.91	-	3,000
Attack Type	DDoS	27,630.92	52.04	531
Industry	IT	24,809.83	51.90	478
Country	UK	16,502.99	51.41	321

Financial Loss by Attack Type

Attack Type	Total Loss (Million \$)	Avg. Loss (Million \$)	Incidents
DDoS	27,630.92	52.04	531
Phishing	26,693.29	50.46	529
SQL Injection	25,156.56	50.01	503
Ransomware	24,479.32	49.65	493
Malware	23,967.95	49.42	485
Man-in-the-Middle	23,550.87	51.31	459

Financial Loss by Industry

Industry	Total Loss (Million \$)	Avg. Loss (Million \$)	Incidents
IT	24,809.83	51.90	478
Banking	22,772.39	51.17	445
Government	21,205.33	52.62	403
Retail	21,119.55	49.93	423
Healthcare	21,041.29	49.05	429
Telecommunications	20,459.09	50.77	403
Education	20,071.43	47.90	419

Financial Loss by Country - Top 10

Country	Total Loss (Million \$)	Avg. Loss (Million \$)	Incidents
UK	16,502.99	51.41	321
Germany	15,793.24	54.27	291
Brazil	15,782.62	50.91	310
Australia	15,403.00	51.86	297
Japan	15,197.34	49.83	305
France	14,972.28	49.09	305
USA	14,812.12	51.61	287
Russia	14,734.73	49.95	295
India	14,566.12	47.29	308
China	13,714.47	48.81	281

Finding	Insight
Highest-loss attack type	DDoS attacks generated the largest financial losses (\$27.63B).
Most targeted industry	The IT sector experienced the highest cumulative losses (\$24.81B).
Country with greatest losses	The UK recorded the highest financial impact (\$16.50B).
Highest average loss per industry	Government organizations suffered the highest average loss per incident (\$52.62M).
Highest average loss per country	Germany had the highest average loss per incident (\$54.27M).

5.3.4 User Impact Analysis

Analysis Performed

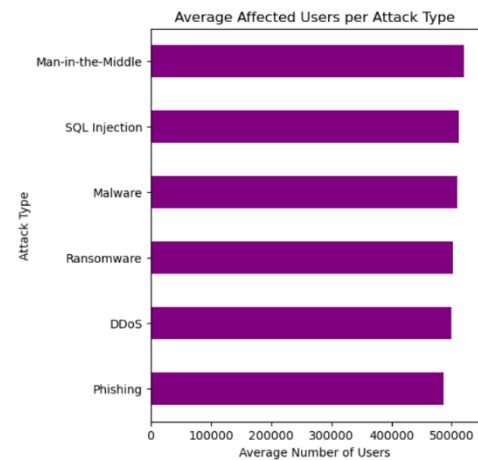
This analysis examined the impact of cyberattacks on users. The analysis included calculating the total number of affected users, measuring the average number of users affected by each attack type, identifying the attack type with the highest user impact, and comparing user impact across industries.

Visualizations

The following visualizations were created to support the analysis:

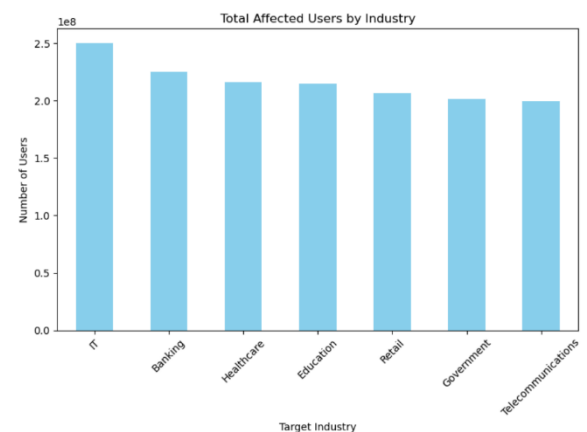
Average Affected Users per Attack Type

A bar chart showing the average number of users affected by each attack type.



Total Affected Users by Industry

A bar chart comparing the total number of affected users across industries.



Results

The analysis shows that **Man-in-the-Middle attacks** have the highest average user impact, affecting approximately **520,064 users per incident**. This makes them the most damaging attack type in terms of the number of users affected.

DDoS attacks also have a very high user impact, affecting around **500,000 users per incident**, followed by **Phishing**, **Ransomware**, **SQL Injection**, and **Malware** attacks.

At the industry level, the **IT sector** recorded the highest user exposure, with approximately **250 million affected users**. The **Banking** and **Healthcare** sectors also experienced a large number of affected users, showing that cyberattacks can impact millions of customers and service users.

Although **Phishing** attacks occur frequently, their average user impact is lower than that of **DDoS** and **Man-in-the-Middle** attacks.

Insight

The findings indicate that industries with large digital infrastructures and customer bases, such as **IT**, **Banking**, and **Healthcare**, face the greatest risk of user disruption. Large-scale attacks on

these industries can affect millions of people and lead to service interruptions, data exposure, and loss of customer trust.

Organizations should focus on improving user protection, strengthening security controls, and developing effective incident response plans to reduce the impact of cyberattacks on customers and users.

5.3.5 Vulnerability Analysis

Analysis Performed

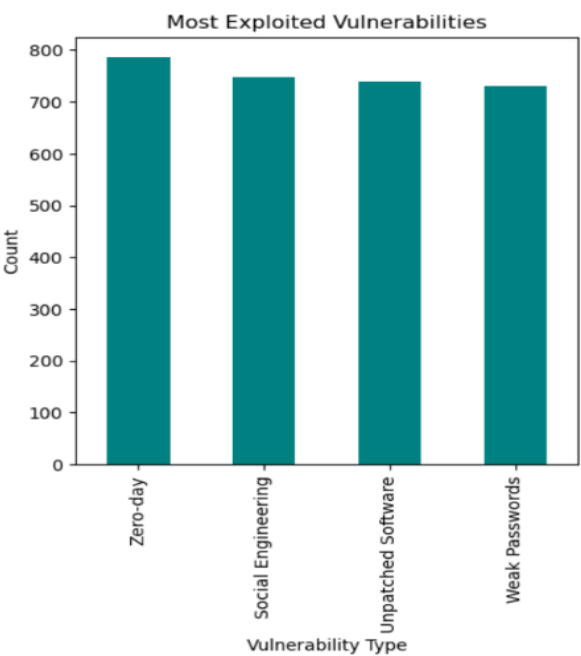
This analysis identified the most exploited security vulnerabilities in the dataset. It also examined how vulnerabilities vary across attack types and industries.

Visualizations

The following visualizations were created to support the analysis:

Most Exploited Vulnerabilities

A bar chart showing the frequency of different types of vulnerability.



Vulnerabilities by Attack Type

A table comparing vulnerabilities used in different attack types.

Attack Type	Social Engineering	Unpatched Software	Weak Passwords	Zero-day
DDoS	132	127	146	126
Malware	130	107	113	135
Man-in-the-Middle	104	135	102	118
Phishing	124	121	124	160
Ransomware	141	124	116	112
SQL Injection	116	124	129	134

Vulnerabilities by Industry

A table showing vulnerability patterns across industries.

Results

The analysis shows that **Zero-day vulnerabilities** are the most exploited weakness, with **785 occurrences**. They are followed by **Social Engineering** (747 occurrences) and **Unpatched Software** (738 occurrences).

Industry	Social Engineering	Unpatched Software	Weak Passwords	Zero-day
Banking	117	107	111	110
Education	115	101	94	109
Government	96	97	98	112
Healthcare	112	105	115	97
IT	107	112	122	137
Retail	97	109	102	115
Telecommunications	103	107	88	105

When examining attack types:

- **Phishing** attacks most frequently exploit **Zero-day vulnerabilities**, with 160 recorded cases.
- **Ransomware** attacks rely heavily on **Social Engineering**, with 141 cases.
- **Man-in-the-Middle** attacks most often exploit **Unpatched Software**, with 135 cases.

When examining industries:

- The **IT sector** is most affected by **Zero-day vulnerabilities**, with 137 incidents.
- The **Healthcare sector** experiences the highest number of attacks involving **Weak Passwords**, with 115 cases.
- The **Government sector** also shows significant exposure to **Zero-day vulnerabilities**, with 112 cases.

The results further show that **Weak Passwords** and **Unpatched Software** remain common security weaknesses across many industries.

Insight

The analysis highlights that organizations continue to face risks from both advanced and basic security weaknesses. While **Zero-day vulnerabilities** are the most exploited, many attacks still succeed because of simple issues such as **weak passwords** and **outdated software**.

The findings suggest that organizations should prioritize regular software updates, strong password policies, employee security awareness training, and vulnerability management programs. Addressing these common weaknesses can significantly reduce the likelihood of successful cyberattacks.

Overall, the results show that improving basic cybersecurity practices remains one of the most effective ways to strengthen organizational security and reduce cyber risk.

5.3.6 Attack Source Analysis

Analysis Performed

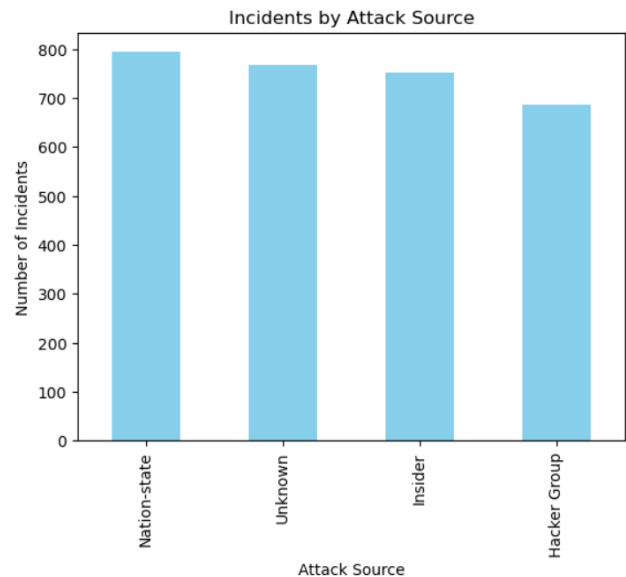
This analysis examined the sources of cybersecurity incidents in the dataset. The analysis included counting the number of incidents by attack source, comparing incident resolution times, and reviewing the relationship between attack sources and cybersecurity risks.

Visualizations

The following visualizations were created to support the analysis:

Incidents by Attack Source

A bar chart showing the number of incidents attributed to Nation-state actors, Unknown sources, Insider threats, and Hacker Groups.

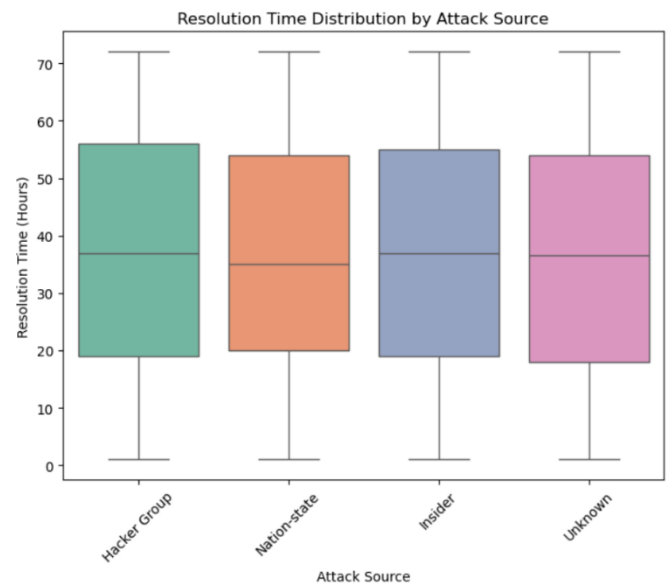


Resolution Time Distribution by Attack Source

A box plot showing the range and distribution of incident resolution times for each attack source.

Results

- **Nation-state actors** are the most common attack source, with around 800 incidents.
- **Unknown sources** are the second largest group, showing the difficulty of identifying attackers.
- **Insider threats** account for about 760 incidents, highlighting the importance of internal security.
- **Hacker Groups** also contribute a significant number of attacks, with around 700 incidents.
- The average incident resolution time is similar across all attack sources, at approximately 35–37 hours.



Insight

The results show that **Nation-state actors** are a major cybersecurity threat due to their high number of attacks and potential financial impact. **Insider threats** remain an important concern because they can be difficult to detect and investigate. The large number of incidents from **Unknown sources** also highlights the challenge of cyberattack attribution. Organizations should strengthen both internal and external security controls to reduce risks and improve incident response.

5.3.7 Defense Mechanism Effectiveness

Analysis Performed

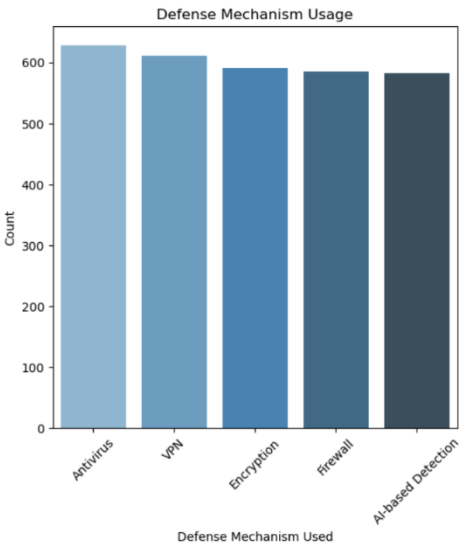
This analysis evaluated the effectiveness of different cybersecurity defense mechanisms. The analysis included measuring how often each defense mechanism was used and comparing incident resolution times across different security controls.

Visualizations

The following visualizations were created to support the analysis:

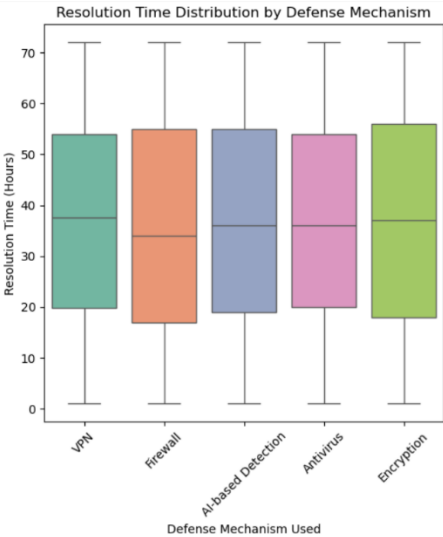
Defense Mechanism Usage

A bar chart showing the frequency of Antivirus, VPN, Encryption, Firewall, and AI-based Detection.



Resolution Time Distribution by Defense Mechanism

A box plot comparing incident resolution times for different security controls.



Results

- **Antivirus** is the most commonly used defense mechanism in the dataset.
- **VPN, Firewall, and Encryption** are also widely used across organizations.
- **AI-based Detection** shows shorter and more consistent incident resolution times than traditional security controls.
- Organizations relying mainly on **VPNs** and **Firewalls** tend to experience longer and more variable resolution times.

Insights

The results suggest that traditional security tools such as Antivirus, VPNs, and Firewalls remain important for basic protection. However, they may not be enough to handle modern cyber threats effectively.

AI-based Detection performs better by helping organizations detect and respond to incidents more quickly. This shows that advanced security technologies can improve incident response efficiency and reduce the impact of cyberattacks.

Overall, organizations should combine traditional security controls with advanced detection technologies to strengthen their cybersecurity defenses.

5.3.8 Incident Response Metrics

Analysis Performed

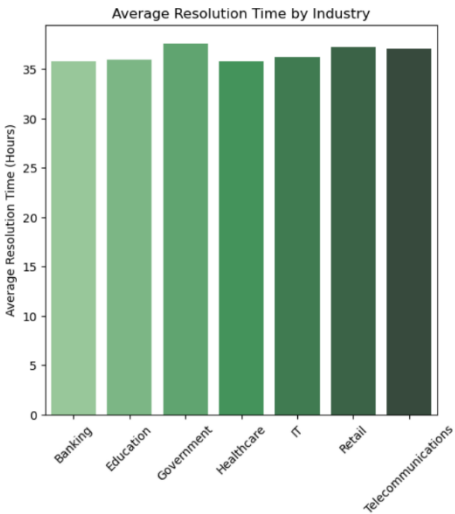
This analysis examined how quickly organizations respond to and resolve cybersecurity incidents. The analysis compared average resolution times across industries and attack types.

Visualizations

The following visualizations were created to support the analysis:

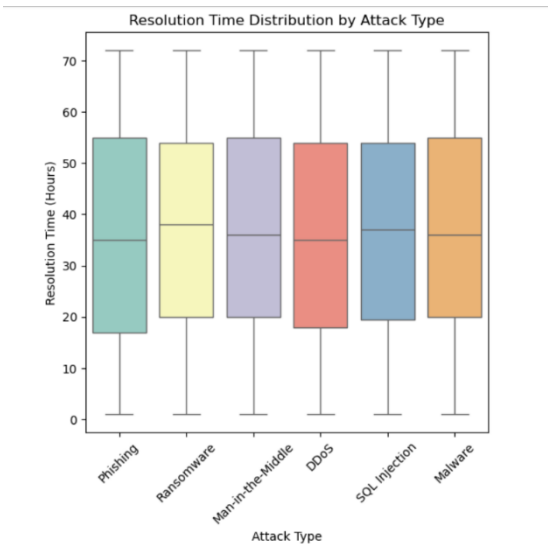
Average Resolution Time by Industry

A bar chart showing the average number of hours required to resolve incidents in each industry.



Resolution Time Distribution by Attack Type

A box plot comparing incident resolution times across different attack types.



Results

- The **Government sector** has the longest average incident resolution time, at approximately **38 hours**.
- Other industries, including **IT, Banking, and Healthcare**, have similar resolution times ranging from **35 to 37 hours**.
- **Ransomware** and **SQL Injection** incidents generally take longer to resolve than other attack types.
- **Phishing** and **DDoS** attacks usually have shorter recovery times.
- Resolution times vary widely, with some incidents resolved quickly and others taking more than **70 hours**.

Insights

The findings show that most organizations require between **35 and 40 hours** to resolve cybersecurity incidents. However, more complex attacks, such as ransomware and SQL injection, often require additional investigation and recovery efforts.

The longer resolution times in the Government sector may be due to the complexity of systems and the sensitivity of the data involved.

Overall, improving detection capabilities, response procedures, and incident management processes can help organizations reduce recovery times and limit the impact of cyberattacks.

5.3.9 Trend Analysis (2015–2024)

Analysis Performed

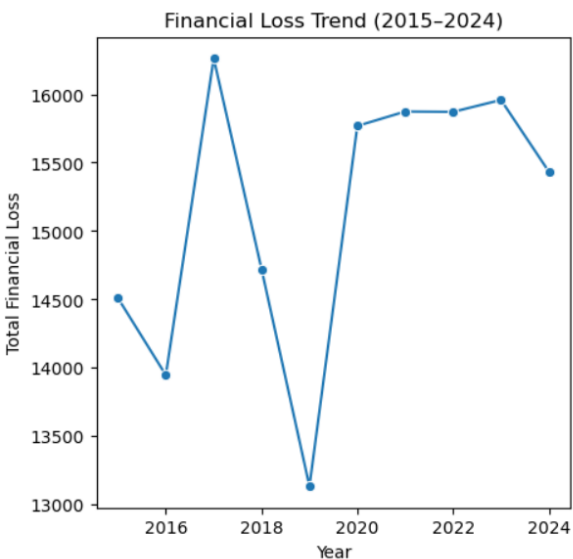
This analysis examined cybersecurity trends between 2015 and 2024. It included tracking financial losses over time, analyzing changes in average incident resolution times, and studying how different attack types changed throughout the years.

Visualizations

The following visualizations were created to support the analysis:

Financial Loss Trend (2015–2024)

A line chart showing changes in total financial losses over time.



Average Resolution Time Trend (2015–2024)

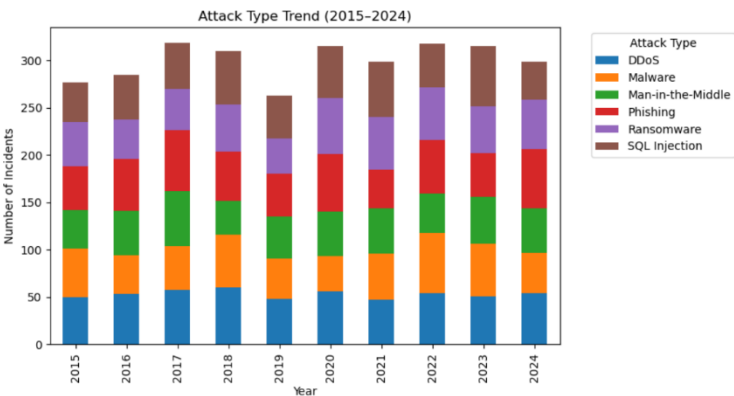
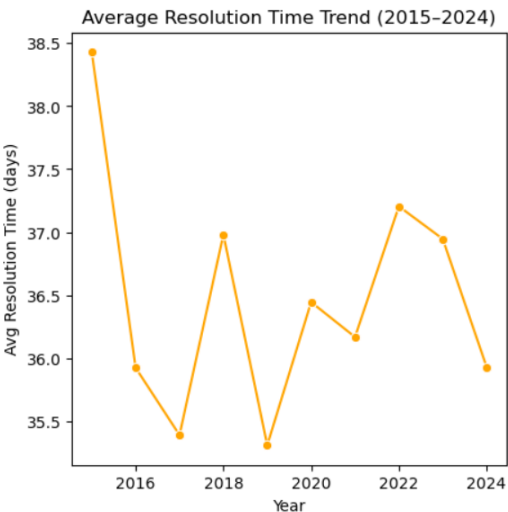
A line chart showing changes in average incident resolution times by year.

Attack Type Trend (2015–2024)

A stacked bar chart showing the distribution of attack types across different years.

Results

- Financial losses increased over the study period, with major peaks occurring around **2017** and **2022**.
- A noticeable decline in financial losses was observed in **2019**.
- Average incident resolution times fluctuated between approximately **35.5** and **38.5** hours.
- Resolution times improved in some years, such as **2017** and **2019**, but increased again in other years.
- **Phishing** and **Ransomware** attacks showed a steady increase over time.
- **DDoS** and **SQL Injection** attacks displayed more variation from year to year.



Insights

The results show that cyberattacks are becoming more costly and continue to pose a growing risk to organizations. The increasing number of phishing and ransomware attacks suggests that these threats are becoming more common and more important for organizations to address.

Although incident response times have shown some improvement, they remain relatively stable overall, indicating that organizations still face challenges in detecting and recovering from cyber incidents.

Overall, the long-term trend highlights the need for stronger cybersecurity strategies, better threat detection, and faster incident response capabilities.

5.3.10 Correlation Analysis

Analysis Performed

This analysis examined the relationships between three important cybersecurity metrics:

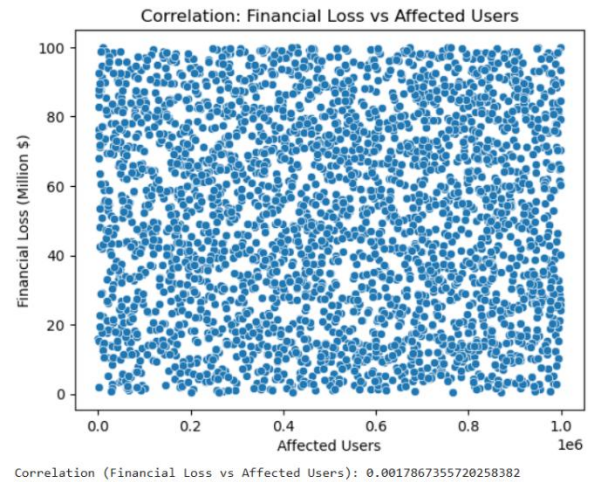
- Financial Loss and Affected Users
- Resolution Time and Financial Loss
- Affected Users and Resolution Time

Visualizations

The following visualizations were created to support the analysis:

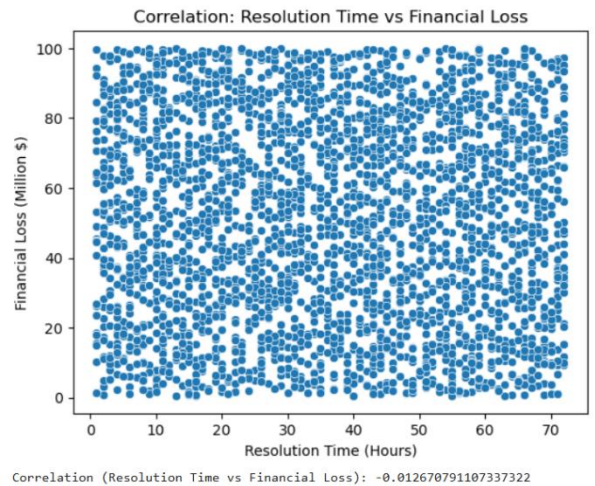
Financial Loss vs Affected Users

A scatter plot showing the relationship between financial loss and user impact.



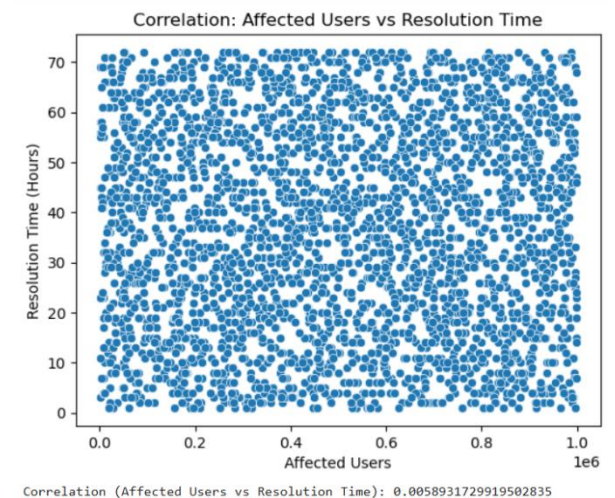
Resolution Time vs Financial Loss

A scatter plot showing the relationship between resolution time and financial loss.



Affected Users vs Resolution Time

A scatter plot showing the relationship between user impact and resolution time.



Results

- The correlation between **Financial Loss** and **Affected Users** is approximately **0.0018**, indicating almost no relationship.
- The correlation between **Resolution Time** and **Financial Loss** is approximately **-0.0127**, showing a very weak negative relationship.
- The correlation between **Affected Users** and **Resolution Time** is approximately **0.0059**, indicating almost no relationship.

Insights

The analysis shows that financial loss, user impact, and incident resolution time are largely independent of each other.

A cyberattack that affects many users does not necessarily cause higher financial losses. Similarly, incidents that take longer to resolve do not always result in greater financial damage.

These findings suggest that cybersecurity incidents should be evaluated using multiple measures rather than relying on a single metric. Financial impact, operational disruption, and user impact can vary significantly depending on the type and complexity of the attack.

Overall, organizations should consider all aspects of an incident when assessing risk and planning cybersecurity strategies.

5.4 Visualization and Dashboard

Purpose

The purpose of the dashboard is to transform cybersecurity data into clear and useful insights. By combining charts, tables, and key performance indicators (KPIs) in one place, users can quickly understand cybersecurity trends, financial impacts, and incident response performance.

Dashboard Components

1. Executive Summary

This section provides an overview of the most important metrics:

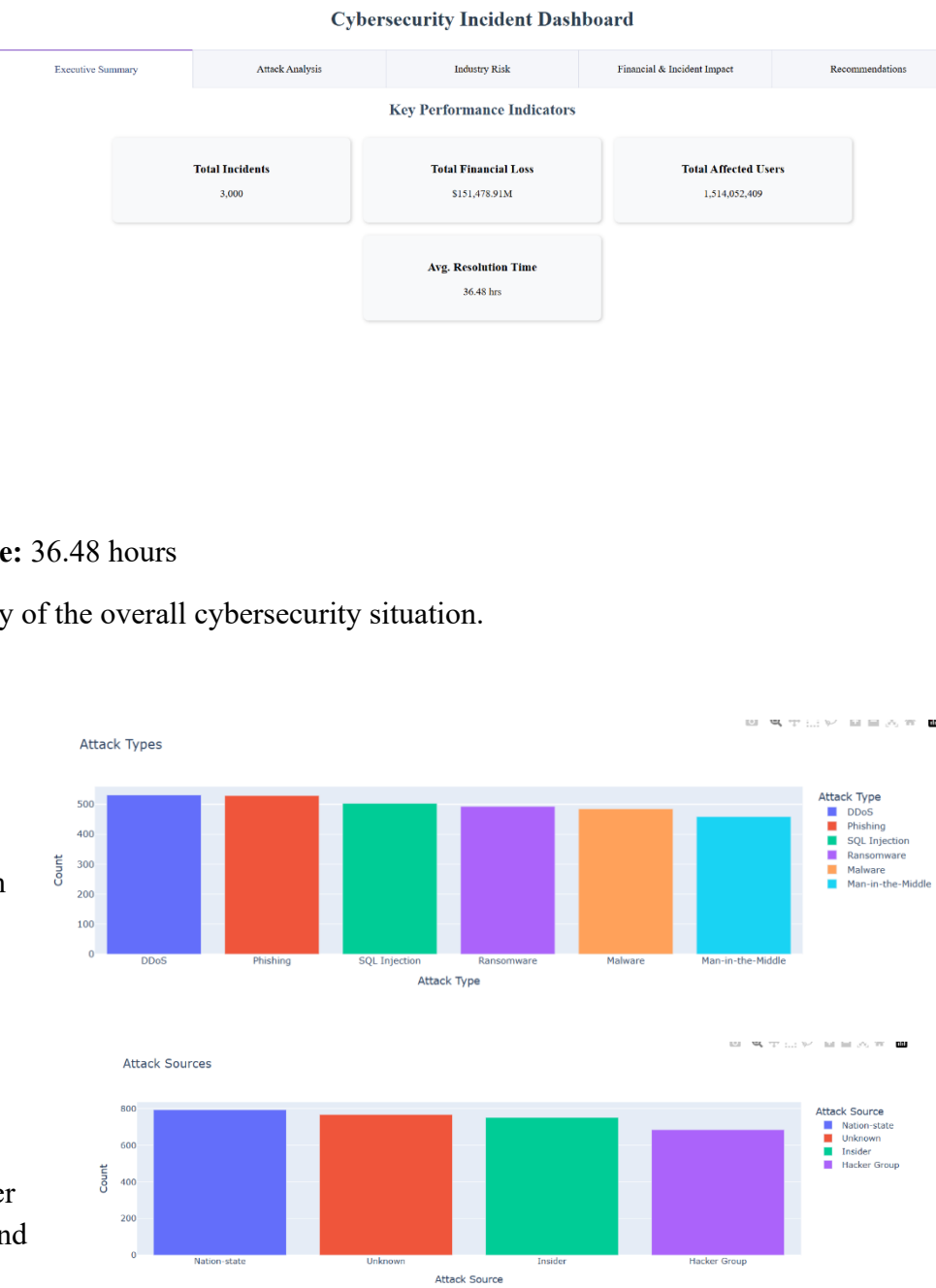
- **Total Incidents:** 3,000
- **Total Financial Loss:** \$151,478.91 million
- **Total Affected Users:** 1.51 billion
- **Average Resolution Time:** 36.48 hours

These KPIs give a quick summary of the overall cybersecurity situation.

2. Attack Analysis

This section focuses on attack types and attack sources.

- Bar charts compare different attack types such as DDoS, Phishing, Ransomware, Malware, SQL Injection, and Man-in-the-Middle attacks.
- Additional charts show attack sources, including Nation-state actors, Insider threats, Hacker Groups, and Unknown sources.



3. Industry Risk Analysis

This section highlights the industries most affected by cyberattacks.

- Bar charts show the number of attacks across industries such as IT, Banking, Healthcare, Government, Retail, Education, and Telecommunications.
- Additional charts compare financial losses and affected users across industries.



4. Financial and Incident Impact Analysis

This section examines the financial and operational impact of cyberattacks.

- Line charts show financial loss trends from 2015 to 2024.
- Line charts display changes in incident resolution times over the same period.
- Scatter plots explore relationships between financial loss, affected users, and resolution time.



5. Recommendations

This section summarizes the main findings and provides recommendations to improve cybersecurity practices and reduce organizational risk.

Results

- The dashboard provides a clear view of cybersecurity trends and risks.
- The **IT sector** shows the highest number of attacks, financial losses, and affected users.
- **Nation-state actors** are the most common attack source.
- Financial losses have increased over time, especially in high-risk industries.
- Incident resolution times remain relatively stable, averaging around 36 hours.
- Correlation analysis shows that financial loss, user impact, and resolution time are not strongly related.

Insights

The dashboard makes complex cybersecurity data easier to understand and analyze. Visualizations help users quickly identify high-risk industries, common attack types, and important trends.

The results show that organizations in the **IT, Banking, Healthcare, and Government** sectors face the greatest cybersecurity risks and should prioritize security investments. The growing financial impact of cyberattacks also highlights the importance of stronger security controls and faster incident response capabilities.

Overall, the dashboard supports data-driven decision-making by helping organizations identify risks, monitor performance, and improve their cybersecurity strategies.

Conclusion

This project analyzed global cybersecurity incidents from 2015 to 2024 to identify attack trends, industry risks, financial impacts, vulnerabilities, and incident response performance.

The results show that the **IT** and **Banking** sectors face the highest cybersecurity risks, experiencing the greatest financial losses and the largest number of affected users. **Nation-state actors** are the most common attack source, while **insider threats** continue to be an important security concern for organizations.

The analysis also found that **Ransomware** and **Phishing** attacks have increased over time and remain two of the most significant cyber threats. In addition, **Zero-day vulnerabilities** are the most frequently exploited security weakness, highlighting the importance of regular updates and vulnerability management.

Although incident resolution times remained relatively stable at around **35 to 38 hours**, more complex attacks such as ransomware often required longer recovery periods. The correlation

analysis further showed that financial loss, user impact, and resolution time are not strongly related. Therefore, organizations should evaluate cybersecurity incidents using multiple measures rather than relying on a single metric.

Overall, the findings demonstrate that cyberattacks continue to grow in both frequency and impact. Organizations must strengthen their cybersecurity strategies to reduce risks, protect sensitive information, and improve incident response capabilities.

Recommendations

1. Strengthen Vulnerability Management

Organizations should regularly update software, apply security patches, and perform vulnerability assessments. Special attention should be given to protecting against Zero-day vulnerabilities through proactive monitoring and threat intelligence.

2. Improve Identity and Access Management

Strong password policies and Multi-Factor Authentication (MFA) should be implemented to reduce unauthorized access. Organizations should also use Privileged Access Management (PAM) to better control access to critical systems and reduce insider risks.

3. Invest in Advanced Security Technologies

Organizations should adopt AI-based detection systems to improve threat detection and reduce incident resolution times. Security Information and Event Management (SIEM) solutions can also help automate monitoring and incident response processes.

4. Apply Industry-Specific Security Measures

Different industries face different risks and should adopt security measures based on their needs.

- **IT and Banking:** Focus on data encryption, ransomware protection, and DDoS mitigation.
- **Healthcare:** Strengthen user authentication and protect sensitive patient information.
- **Government:** Improve incident response planning and insider threat monitoring.

5. Enhance Incident Response Processes

Organizations should develop incident response playbooks for common threats such as ransomware and phishing attacks. Regular security exercises and simulations can help teams prepare for real-world incidents and improve response efficiency.

6. Increase Security Awareness

Employees should receive regular cybersecurity training to help them recognize phishing attacks and social engineering techniques. In addition, executive dashboards should be used to provide management with clear visibility into cybersecurity risks and performance metrics.

Final Recommendation

Organizations should combine strong security controls, advanced detection technologies, effective incident response processes, and employee awareness programs to build a stronger cybersecurity posture. A proactive and data-driven approach will help reduce cyber risks, minimize financial losses, and improve overall security resilience.